

Developing a Plugin-Based Knowledge-Base Construction Pipeline for Multi-Domain OMNI-RAG: Case Studies in Indonesian Education Regulations and Netgear User Manuals

Moh Fairuz Alauddin Yahya | 13522057 | Program Studi Informatika, Institut Teknologi Bandung

ABSTRACT

Retrieval-augmented generation (RAG) relies on a knowledge base that can retrieve the right evidence before a language model composes an answer. This becomes difficult in a multi-domain setting: Indonesian education regulations and Netgear user manuals differ in hierarchy, terminology, and search behavior. Structure-agnostic chunking can break context or discard source identity, while domain-specific indexing and relation-building logic can be hard to extend.

This work develops and evaluates a plugin-based knowledge-base construction pipeline for OMNI-RAG. Shared canonical artifacts and compatibility contracts decouple the orchestrator from four chunking strategies, three indexing strategies, and two knowledge-graph strategies: a deterministic legal graph and an LLM-based graph. The same canonical chunks feed sparse, dense, or hybrid indexes and an optional legal graph; profiles, manifests, and provenance record component selection and preserve traceability. We evaluate nine Netgear manuals (1,400 pages; 200 bilingual question instances derived from 100 slots) and a legal corpus of 256 documents with 25 verified questions, including 49 reference documents and 207 stratified noise documents. Retrieval is evaluated at $K=30$, with cutoffs swept from $K=5$ to $K=60$, while generation uses the top ten evidence items.

On the manual corpus, structure-aware hybrid chunking obtains Recall@30 of 0.740, and structure-aware dense indexing reaches 0.785; the corresponding generation correctness/faithfulness scores are 0.788/0.958. On the legal corpus, legal-structure dense retrieval reaches 0.6854 and generation correctness/faithfulness scores 0.690/0.972, while hybrid indexing gives the highest legal MRR at 0.5439. Legal-graph traversal to depth two reaches neighborhood recall as high as 1.000, a measure reported separately from context-reference recall. The results show that preserving document structure and combining semantic matching with explicit relations can improve evidence coverage, but the best configuration depends on the domain and the metric prioritized.

Keywords: OMNI-RAG, retrieval-augmented generation, knowledge-base construction, chunking, indexing, knowledge graph, plugin architecture.